IBM Z DATATHON 2026 | PROJECT PITCH

Z-AgentShield

A policy-driven security gateway prototype for enterprise AI agent actions

Let AI act. Keep every action under control.



Team Z Innovators

DRAFT - confirm official PDF naming and submission rules

01 | THE CHALLENGE

Agent actions need a control point

What changes with autonomous agents?

- Agents can propose many tool and data operations at machine speed.
- Individual services may apply different authorization and logging rules.
- High-impact actions need consistent policy decisions and review routing.

What the prototype addresses

- Evaluate proposed actions against ordered enterprise policy rules.
- Record a decision and selected redacted evidence in SQLite.
- Present history, pending reviews, and policy previews in one local console.

Scope: the gateway evaluates actions. It does not execute downstream operations or connect to enterprise systems.

One decision path, visible evidence

01
Intercept
FastAPI validates a request and applies request identity checks.

02
Evaluate
Ordered YAML rules, risk scoring, and optional anomaly analysis inform a decision.

03
Account
SQLite persists decision evidence and pending review tickets; Streamlit displays bounded views.

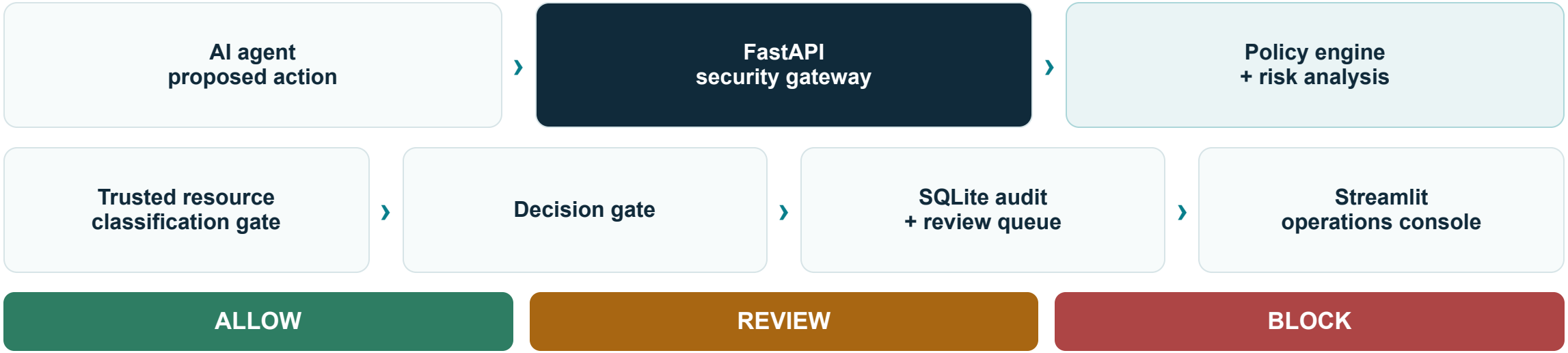
ALLOW
only with trusted target verification

REVIEW
pending human review

BLOCK
mandatory or default deny

The default deployment has no trusted resource classifier; policy ALLOW is therefore converted to BLOCK. This is fail-closed behavior, not a successful authorization claim.

Proposed action to recorded decision



Rule order: mandatory deny → review → explicit allow → default deny. Decisions and selected metadata are recorded; no downstream action is executed by this prototype.

Controls first; limitations shown

Implemented safeguards

- Mandatory deny rules and default-deny policy behavior.
- Policy ALLOW is blocked unless a trusted resource classifier verifies the target.
- Selected sensitive parameter keys and inline values are redacted.
- Swagger/OpenAPI is disabled by default and enabled only in explicit development mode.
- Dashboard review actions remain read-only without trusted reviewer authentication.

Not claimed

- No production identity provider or trusted reviewer authentication.
- No authoritative resource registry/classifier configured by default.
- Redaction is heuristic; SQLite is not cryptographically immutable.
- Event records do not identify synthetic versus operational provenance.
- No IBM service integration or verified LinuxONE deployment.

Evidence from repeatable local checks

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Unit tests

Passed with exit code 0 under PATH Python 3.13.15.

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Live smoke checks

Passed against isolated local API and dashboard instances with disposable SQLite storage.

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Dashboard sections

SOC overview, human review, audit trail, behavioral analytics, policy simulator.

Observed in isolated verification: public-read request BLOCKED by the fail-closed classifier gate; bulk export REVIEW; repeated request ID returned the stored decision; unauthenticated review/audit access returned 401; synthetic secret marker was absent from audit output.

Local dashboard: http://127.0.0.1:8501

Local API docs (development only): http://127.0.0.1:8000/docs

The verification writes only synthetic records to a disposable database, which was removed after the run. HTTP checks do not prove production readiness.

From prototype to a deployable control plane

Next engineering steps

- Integrate a server-controlled resource classifier and test failure behavior.
- Add authenticated reviewer identity, authorization, and auditable decisions.
- Validate audit integrity, retention, privacy, and event provenance requirements.
- Deploy and test on IBM LinuxONE only when access is available; capture genuine evidence.

Submission and project status

- Team: Team Z Innovators.
- Source: `github.com/chakriburidi237-crypto/Z--AgentShield-IBM-2026`
- License: Apache-2.0.
- Official deck filename, remaining submission metadata, and hacking-window eligibility still require confirmation.
- This deck is a factual draft; no demo video or LinuxONE evidence is included.

Ask: help validate the resource trust boundary, reviewer identity integration, and deployment requirements before connecting this prototype to real systems.